ME602T Assignment 6

Name: Dusi Venkata Surya Gowri Nageswar
Roll Number: ME24BT018

---

1. Numerical Method

1.1 Discretisation

With $\Delta x = \Delta y$ and second-order central differences, the interior nodes (i, j) satisfy:

$$T_{i,j} = \frac{1}{4}(T_{i-1,j} + T_{i+1,j} + T_{i,j-1} + T_{i,j+1})$$

1.2 Convective boundary

The top boundary nodes ($j = n$) are treated as unknowns and updated with the same five-point stencil. This needs a ghost node at $j = n + 1$ outside the plate. Applying a central difference to the convective condition at the boundary node:

$$-k\,\frac{T_{i,n+1} - T_{i,n-1}}{2\Delta y} = h\left(T_{i,n} - T_\infty\right)$$

$$\Rightarrow \quad T_{i,n+1} = T_{i,n-1} + \frac{2h\Delta y}{k}\left(T_\infty - T_{i,n}\right)$$

The ghost values are refreshed after every sweep using the latest boundary-node temperatures.

1.3 Given Values

$h = 10 \mathrm{W/m^2.K}$
$k = 50 \mathrm{W/m.K}$
Bottom and Side walls temperature $T_{wall} = 1000K$
Ambient Temperature $T_{inf} = 300K$
Length of Plate $L = 1m$ in both X and Y direction.

1.4 Point Gauss-Seidel (PGS)

Nodes are swept in row-major order, and every update uses the most recent neighbour values:

$$T_{i,j}^{k+1} = \frac{1}{4}\left(T_{i-1,j}^{k+1} + T_{i+1,j}^{k} + T_{i,j-1}^{k+1} + T_{i,j+1}^{k}\right)$$

1.5 Point Successive Over-Relaxation (PSOR)

PSOR blends the old value with the Gauss-Seidel value using a relaxation factor ω:

$$T_{i,j}^{*} = \frac{1}{4}\left(T_{i-1,j}^{\,k+1} + T_{i+1,j}^{\,k} + T_{i,j-1}^{\,k+1} + T_{i,j+1}^{\,k}\right)$$

$$T_{i,j}^{\,k+1} = T_{i,j}^{\,k} + \omega\left(T_{i,j}^{*} - T_{i,j}^{\,k}\right)$$

$\omega = 1$ recovers PGS. $1 < \omega < 2$ is over-relaxation, which is needed for convergence to be accelerated.

1.6 Convergence criterion

After each sweep the RMS of the change in temperature over all updated nodes is computed:

$$e = \sqrt{\frac{1}{N}\sum_{i,j}\left(T_{i,j}^{\,k+1} - T_{i,j}^{\,k}\right)^2}$$

where $N = (m-1) \times n$ is the number of updated nodes. Iteration stops when $e < 10^{-6}$ K. The RMS change underestimates the true error, so a strict tolerance of 10^{-6} K was used.

---

2. Algorithm

1. Set the parameters ($L_x, L_y, m, n, h, k, T_\infty$) and compute Δx and $c = 2h\Delta x/k$.
2. Allocate the temperature array of size $(n+2) \times (m+1)$ (the extra row holds the ghost nodes).
3. **Initialise:** set left, right and bottom boundaries to 1000 K, and all other nodes to the average of the surrounding boundary values, $(3 \times 1000 + 300)/4 = 825$ K.
4. **Iterate** until $e < 10^{-6}$:
   Sweep $j = 1 \ldots n$ and $i = 1 \ldots m-1$.
   - PGS: $T_{i,j} \leftarrow \frac{1}{4}(4 \text{ neighbours})$.
   - PSOR: $T_{i,j} \leftarrow T_{i,j} + \omega(T^{*} - T_{i,j})$.
     Accumulate the squared change and compute e.
     Update the ghost row: $T_{i,n+1} = T_{i,n-1} + c\,(T_\infty - T_{i,n})$.
     Increment the iteration counter.
5. **PGS:** write the full field and the profiles at $x = 0, 0.25, 0.5$ m ($i = 0, 100, 200$) to files.
6. **PSOR:** repeat steps 3–4 for $\omega = 1.1, 1.2, \ldots, 1.9$, and record iterations against ω.
7. Identify the ω with the fewest iterations and compare with PGS.

---

3. Results

3.1 Point Gauss-Seidel

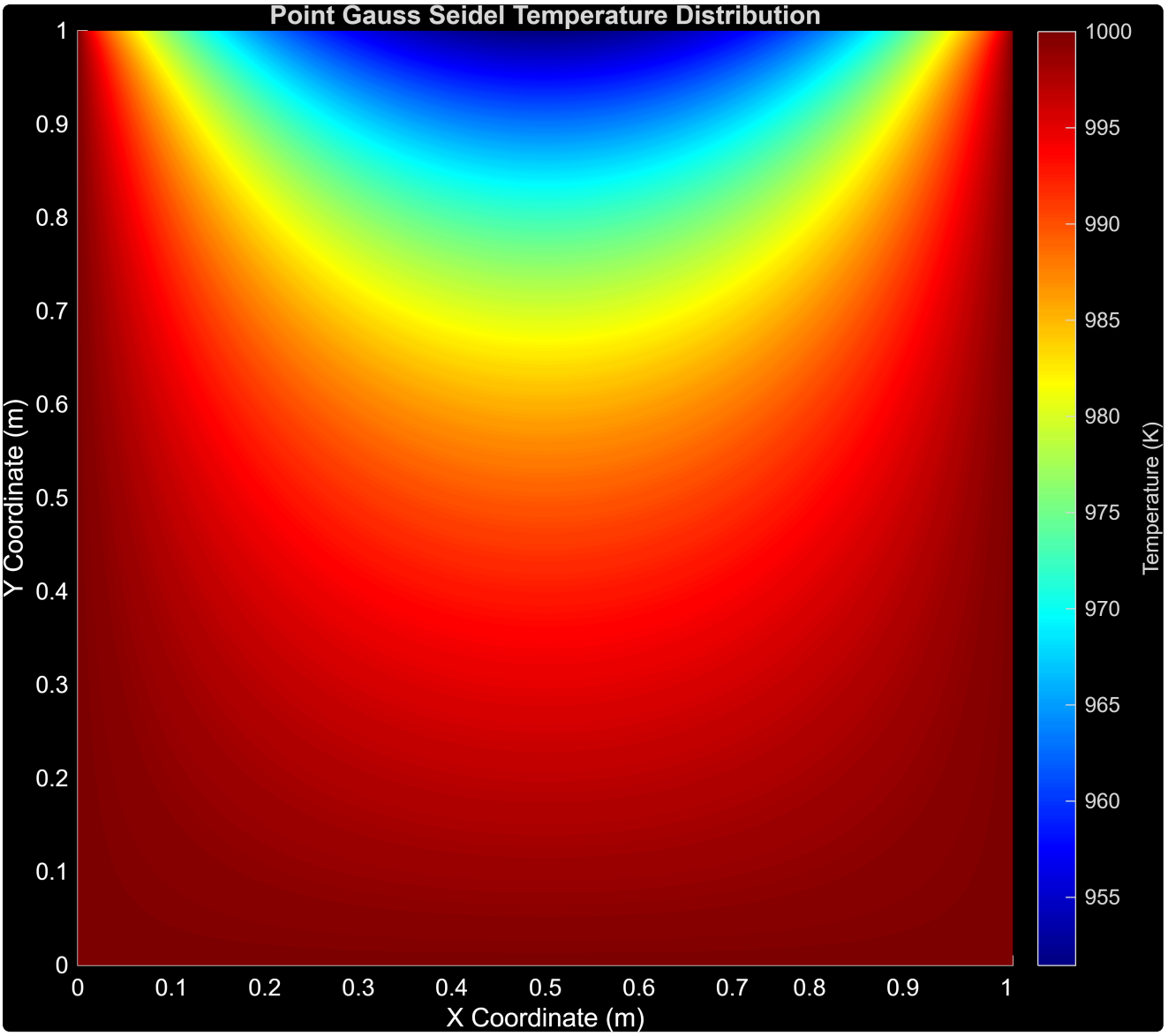


Figure 1: Steady-state temperature contour (PGS).

Quantity	Value
Iterations to converge	215134

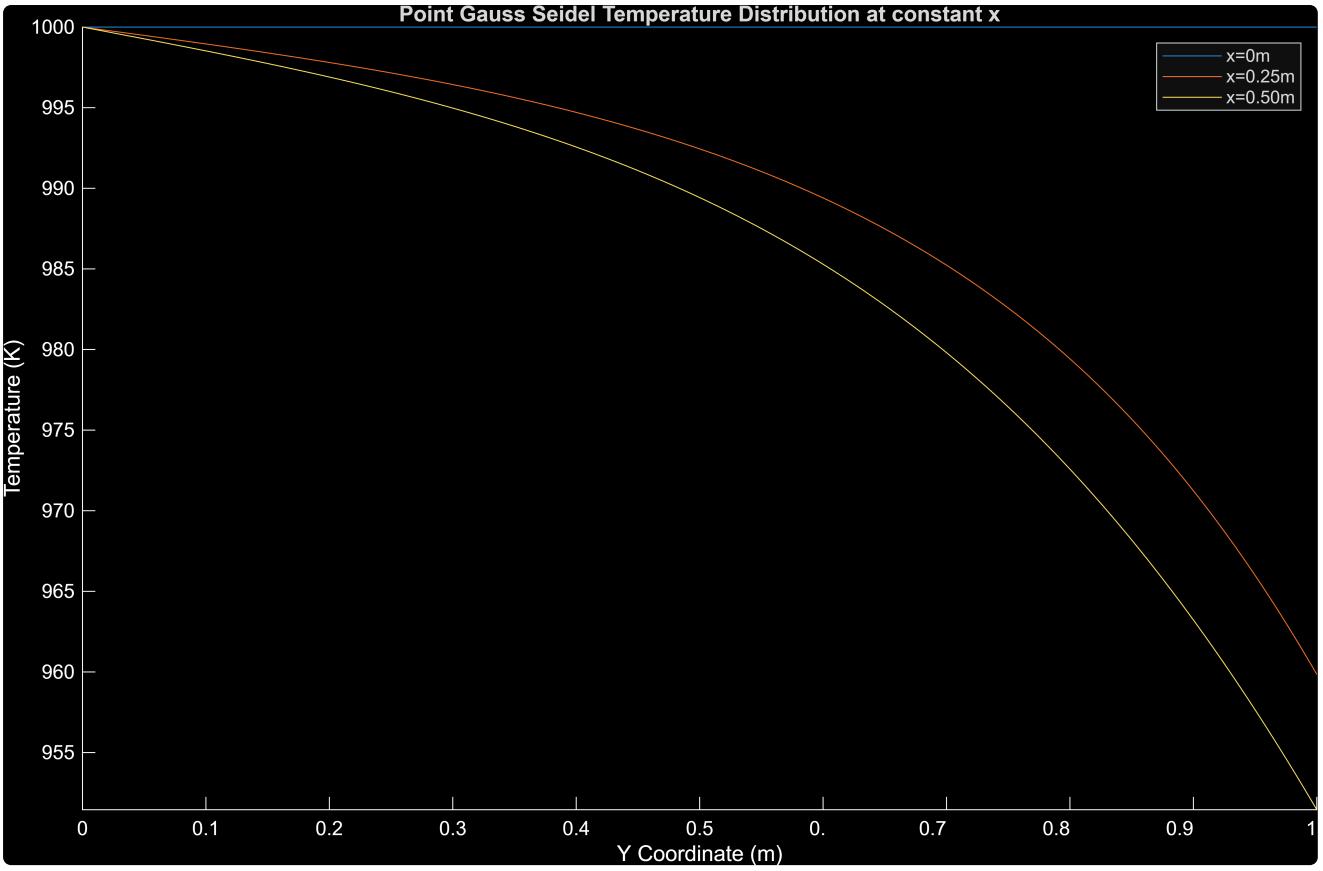


Figure 2: Temperature distribution along y at $x = 0.0$, 0.25 and 0.5 m (PGS).

Observations

- At $x = 0$ the profile is flat at 1000 K because it lies on the left wall.
- At $x = 0.25$ and 0.5 m, the temperature falls towards the top boundary, with the lowest value at $y = 1$ m where heat is lost by convection.
- Along $x = 0.5$ m (the centreline) the drop at the top is the largest, since it is furthest from the 1000 K side walls.

3.2 Point Successive Over-Relaxation

ω	**Iterations**
1.1	180183
1.2	150286
1.3	124315
1.4	101447
1.5	81057
1.6	62655
1.7	45838
1.8	30236
1.9	15426

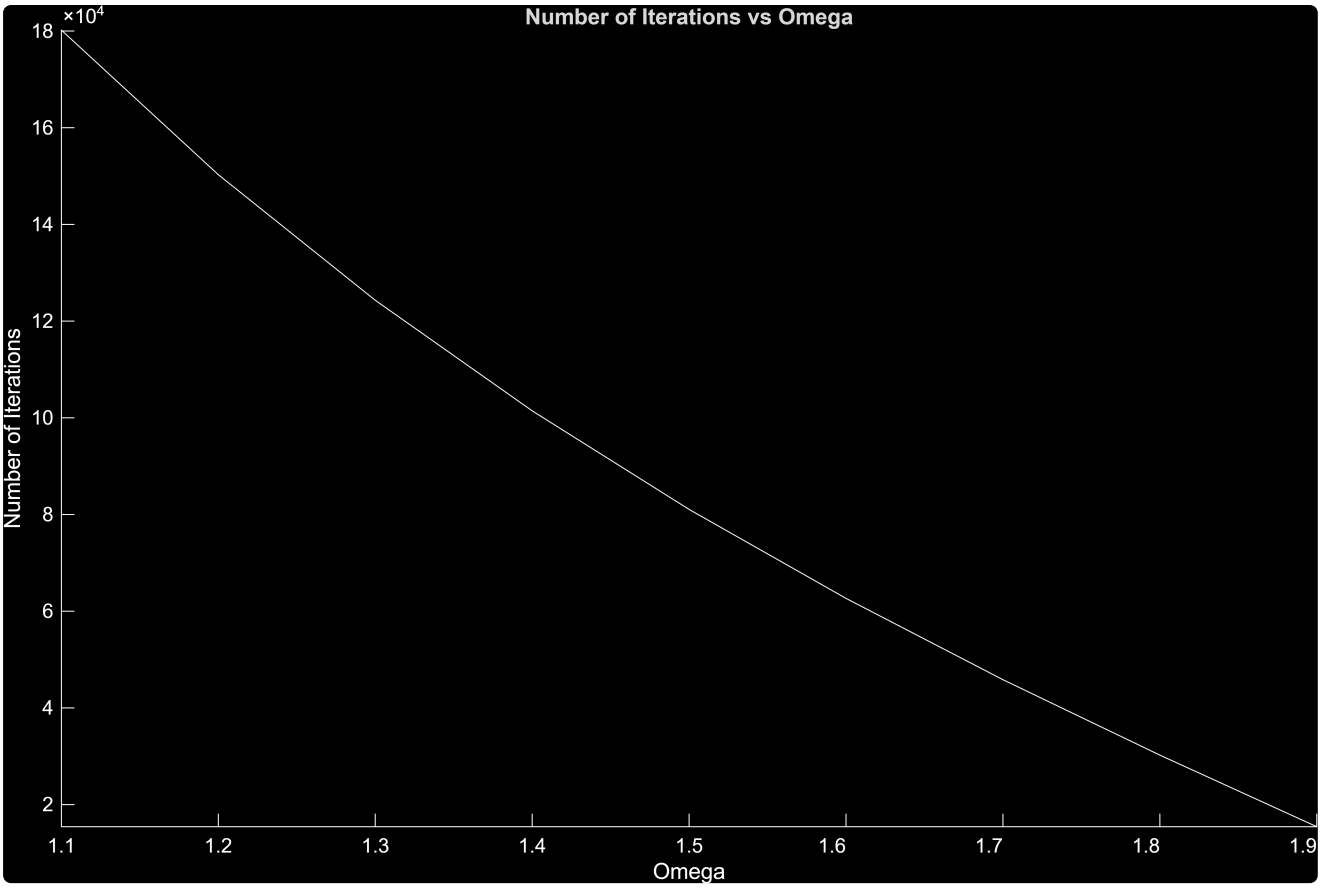


Figure 3: Number of iterations vs. relaxation factor ω.

Optimal relaxation factor: ω_{best} = 1.9, requiring 15426 iterations.

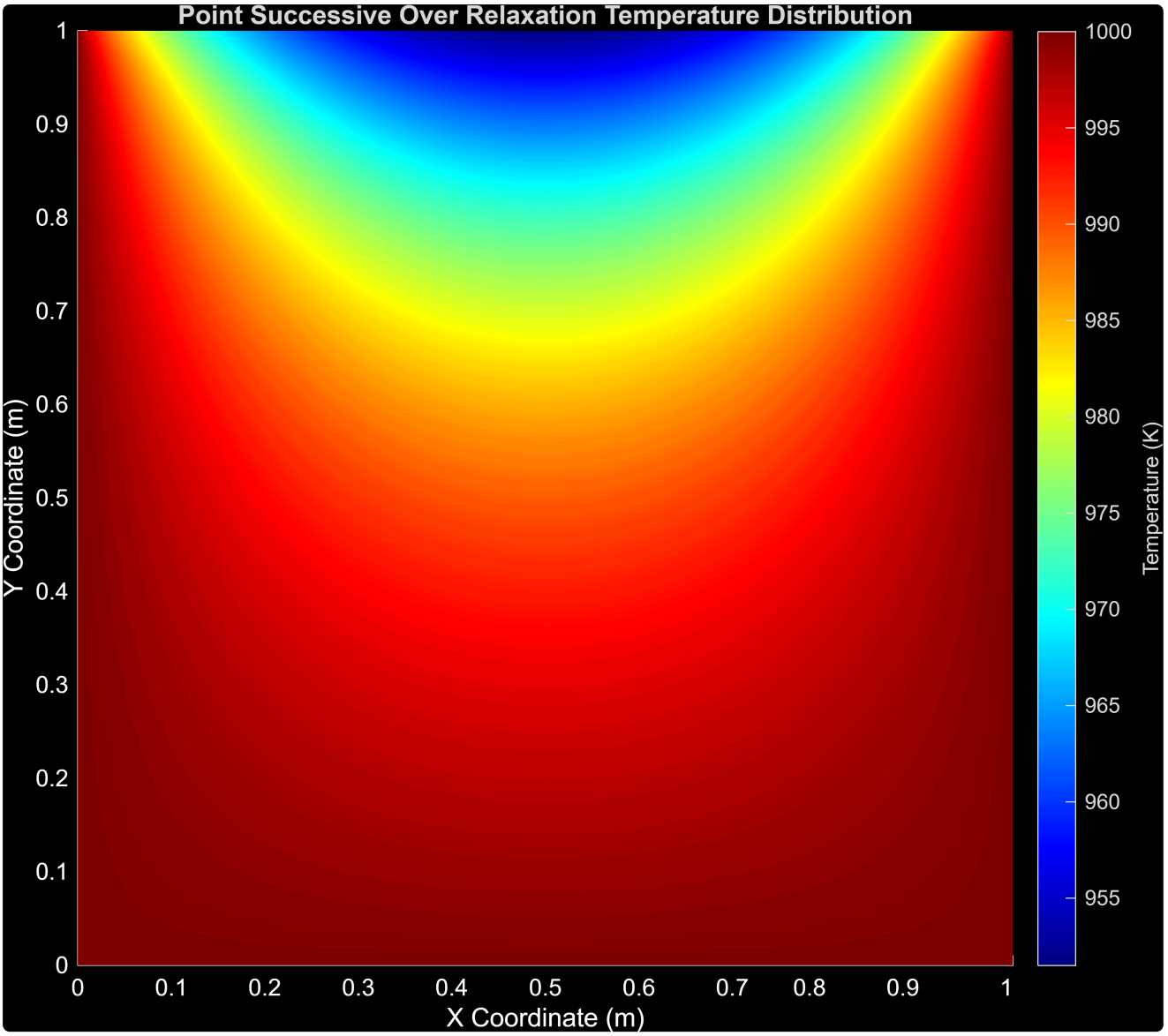


Figure 4: Temperature contour from PSOR at ω_{opt}.

3.3 Comparison of methods

Method	Least iterations
PGS ($\omega = 1$)	215134
PSOR ($\omega = 1.9$)	15426

---

4. Discussion

- PSOR with $\omega > 1$ converges much faster than PGS because over-relaxation extrapolates each update along the direction of the Gauss-Seidel correction.
- The iteration count falls as ω increases towards 2 and rises sharply beyond the optimum (divergence for $\omega \geq 2$). Sharp rise cannot be captured as ω_{opt} is very close to 2.
- From observed values ω_{best} is obtained as 1.9

- The convective top boundary lowers the temperature near $y = 1$ m, and the effect is strongest at the center, away from the hot side walls.

5. Conclusion

PGS needed 215134 iterations, while PSOR at $\omega_{best} = 1.9$ needed only 15426 iterations, a reduction of about 92.8%. Over-relaxation is therefore a highly effective and cheap way to accelerate point-iterative solvers.